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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COS 201 — Past Questions & Materials

by Renance

Course: COS 201

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 200

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Paper 1 of 1 — COS 201 Computer Programming 2024/2025

Level 200 • First Semester • PastQuestion

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DEPARTMENT OF COMPUTER SCIENCE
COLLEGE OF SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

2024/2025 FIRST SEMESTER EXAMINATION

COS 201 [Computer Programming I]

INSTRUCTIONS

- Answer Four questions
- Time Allowed: 2Hrs 20mins
- Do not mix up questions. Start a new question on a fresh page.
- All extra sheets used should be signed by the Invigilator

QUESTION ONE

- Write a program that solves for the real roots of the quadratic equation $ax^2 + bx + c = 0$. Using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. 10mks
- Write a program to calculate the area of a circle using the formula $A = \pi r^2$, given a numeric value for the radius r as input data. 4mks
- Write a program to print the first 10 multiples of 5. 4mks

QUESTION TWO

- A common problem in personal finance is that of determining how much money will accumulate in a bank account after n years if a known amount, P , is deposited initially and the account collects interest of a rate r percent per year compounded annually. Write a program for this compound interest using the formula $A = P \left(1 + \frac{r}{n}\right)^{nt}$. 12mks
- Calculate the volume and area of a sphere using the formula $S = 4\pi r^2$, $V = \frac{4}{3}\pi r^3$. 6mks

QUESTION THREE

- Write a loop that will calculate the sum of every third integer, beginning with $i=2$ (i.e., calculate the sum $2 + 5 + 8 + 11 + \dots$) for all values of i that are less than 100. Write the loop in three different ways (i) Using a *while* statement (ii) Using a *do-while* statement (iii) Using a *for* statement. 12mks
- Using an *if* statement, write a program that will input integers that can be divided by both 2 and 3. 6mks

QUESTION FOUR

- Write a program that prints the *perimeter* of a rectangle given its *height* and *width*. ($Perimeter = 2 * (width + height)$). 6mks
- Using a *while* statement, write a program to print the first 10 natural numbers. 6mks
- Write a program to reverse the digits of a given number. 6mks

QUESTION FIVE

- Write a program that converts 27° from Fahrenheit (F) to degrees Celsius (C) using the formula: $C = (F - 32) / 1.8$. 6mks
- Write a program that gives the number of hours an employee worked and the hourly wage, compute employee's weekly pay for seven workers. 8mks
- Write a program to calculate the absolute value of an integer. 4mks